

Monitoring of landslides in Nainital, Uttarakhand, India: Validation of PS-InSAR results

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Abstract—This paper investigates the efficacy of Stanford Method for Persistent Scatterers (StaMPS) to monitor the critical landslides in Nainital township and adjoining areas of state of Uttarakhand, India which has problems due to inadequate drainage management system, seismically active Main Boundary Thrust (MBT) passing from the area, unwanted construction on highly unstable slopes and low *insitu* strength of rocks. In this research work, 13 descending ENVISAT ASAR C-Band images of Nainital acquired between October 2008 to August 2010 are processed to study the PS-InSAR time series analysis for measuring surface displacement. The case study indicates that StaMPS efficiently extracted sufficient number of PS pixels and detected the slow movement occurring in some regions. In the study area, few critical zones are also identified where time series 1D-Line of Sight (LOS) displacement map shows the maximum deformation (away from the satellite) of about 17 mm/year.

Keywords—PS-InSAR, StaMPS, Landslide, Monitoring

I. INTRODUCTION

SAR Interferometry (InSAR) has been extensively used for deformation monitoring due to capability of imaging earth in all-weather conditions, high resolution and security advantage over conventional monitoring methods such as GPS and levelling [8]. However, due to several error sources such as orbital errors, atmospheric delay, temporal and spatial decorrelation, etc., effectiveness of InSAR observations was always under scrutiny [4]. Therefore, a novel approach called the Persistent Scatterer SAR Interferometry (PS-InSAR) was conceptualized and developed [1]-[2]. The developed approach overcomes the problem of decorrelation by identifying resolution elements with a stable amplitude and phase history over a long time interval and for wide look-angle variations. Due to time series analysis, all the aforementioned error sources limiting the application of InSAR can be estimated and removed by combining SAR images of long time interval. Although PS-InSAR has proven to be more accurate and reliable over the InSAR, this technique too has certain limitations such as inefficiency in obtaining high PS pixel density in rural areas, representation of surface deformation in one-dimensional line of sight (1D-LOS) direction and revisit period of satellite.

Since PS-InSAR has shown its potential in monitoring surface deformation, many researchers have already used this

approach to monitor deformation for land subsidence, volcanic deformation, landslides, etc. [7]-[10]. Presently, two methods namely DePSI, the PS-InSAR package developed at Delft University of Technology (TU Delft) and StaMPS are frequently used for time series analysis. Reference [6] has already compared the above two in study of Granada basin, southern Spain and concluded that in urban areas both techniques produce similar results. However, in non-urban areas such as forests, agriculture or mountains, StaMPS performed better and efficiently extracted more PS pixels.

In this study, we have applied the StaMPS based PS-InSAR approach for monitoring landslide prone areas in Nainital township of Uttarakhand, India. The status of those landslides may be investigated further for stabilization of unstable slopes.

II. STUDY AREA AND SATELLITE DATASET

In the complex geological environment of the Himalaya, landslide is one of the most threatening geo-hazards which causes colossal damages to the urban infrastructure and affects the socio-economic condition of local community. Disaster Management and Mitigation Center [5], Uttarakhand has published a landslide hazard zonation map for the state which has categorized landslide prone areas in three classes from moderate to low, high, and moderate to very high in terms of instability.

Nainital town, in Kumaon Himalaya, a popular tourist destination, is spread around the periphery of a 1.4 km long kidney shaped lake, surrounded by high hill ranges. The township is located in the close vicinity of Main Boundary Thrust (MBT) which is considered as seismically active and thus increases the vulnerability of township. Although in recent times, no major tectonic event has been reported but the area falls in Zone IV of Earthquake Zoning Map of India (IS1893, Part 1, 2002). In addition, there are multiple fault traces observed in Nainital and a major tectonic discontinuity known as lake fault nearly bisects the lake. The Nainital lake (Fig. 1) is bounded by Sher-ka-danda ridge to the east, Naina peak (2610m) to the north; Ayarpatha ridge (2352m) on the west and has a narrow stream outlet called as Balia Nala towards south [11]. There are a few areas of active landslides such as Golf course (near Raj Bhawan), Harinagar-Balia Nala, Krishnapur and Hanumangarhi areas. The DMMC report [5]

mentions that the landslides in Harinagar-Balia Nala and Golf course are in critical state and impression of movements can be clearly observed. A massive landslide at Golf course can be identified to show rapid headward progression with presence of transverse and crown cracks.

The town has experienced disastrous landslide events in 1867, 1880, 1893, 1898, 1924 and 1998. Sher-ka-danda hill was affected by the mass movements in 1860 and 1888 which caused severe damage to infrastructure and cost many lives [11].

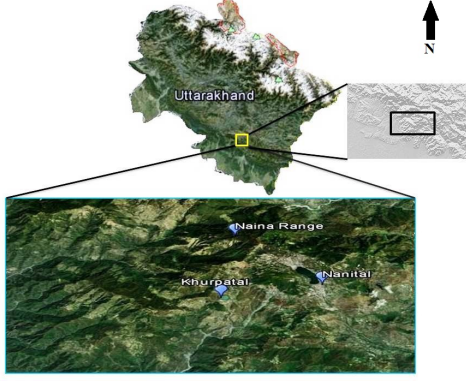


Fig. 1. Google earth image showing study area with Nainital township around lake and its SRTM DEM

To monitor the critical slopes around Nainital lake, we have processed 13 descending ENVISAT C-Band ASAR images of track 248 (frame: 3015) acquired between October 2008 to August 2010. ASAR image of 25th December 2009 is chosen as the master image based on minimizing the temporal, Doppler and perpendicular baseline [4]. The dataset is presented in Table I along with the critical perpendicular baseline length with respect to master image. Apart from the SLC images, a 3 arc second (90 m resolution) SRTM Digital Elevation Model (DEM) is used to remove the topographic phase from the differentially corrected interferogram. The orbital corrections are done with the help of precise orbits obtained from ESA. The DORIS precise orbits for the year 2008, 2009 and 2010 are used to remove orbital bias.

Table I
ENVISAT data for Nainital Area (track: 248; frame: 3015)

S. No.	Acquisition date	Orbit No.	Perpendicular baseline (B) (m)	Doppler Centroid frequency(Hz)
1.	31-OCT-2008	34871	155.6	166.231
2.	05-DEC-2008	35372	43.7	168.060
3.	07-AUG-2009	38879	44.9	170.098
4.	11-SEP-2009	39380	476.3	165.710
5.	16-OCT-2009	39881	-161.1	163.154
6.	20-NOV-2009	40382	377.8	171.735
7.	25-DEC-2009	40883	0	162.437
8.	29-JAN-2010	41384	339.2	171.353
9.	05-MAR-2010	41885	28	168.613
10.	09-APR-2010	42386	324.5	172.944
11.	18-JUN-2010	43388	252.8	182.534
12.	23-JUL-2010	43889	-65.2	177.749
13.	27-AUG-2010	44390	-37.9	177.370

III. STAMPS APPROACH

The StaMPS method is an improvement over the previous PS-InSAR techniques as it is able to identify PS pixels in urban as well as nonurban areas [4]. The method uses phase characteristics of the pixels present in a SAR image to isolate a set of phase stable pixels independent of amplitude, which helps in extracting low amplitude phase pixels as well. The approach also uses spatial correlation of the interferogram phase in a series of interferograms which eradicates the requirement of a deformation model for choosing pixels based on similarity in phase history [3], [4]. The StaMPS method contains four steps: interferogram generation, phase stability estimation, PS selection and displacement estimation.

The image which minimizes the sum decorrelation (ρ_{total}) is chosen as the master image for interferogram formation.

$$\rho_{total} = \rho_{temporal} \cdot \rho_{spatial} \cdot \rho_{doppler} \cdot \rho_{thermal} \approx \left(1 - f\left(\frac{T}{T_c}\right)\right) \cdot \left(1 - f\left(\frac{B_{perp}}{B_{c,perp}}\right)\right) \cdot \left(1 - f\left(\frac{F_{DC}}{F_{DC,c}}\right)\right) \cdot \rho_{thermal} \quad (1)$$

The interferograms are generated by using this master image which involves co-registration, resampling and complex multiplication of the master with all the slave images. Once interferograms are generated, the phase stability test is applied to obtain a set of phase stable pixels. An initial amplitude based filtering is performed using the amplitude dispersion index ' D_A ' value which is the ratio of the standard deviation (σ_A) and the mean (μ_A) of the amplitude values to filter the pixels which are not PS candidates [4]. After selecting a subset of PS candidates, phase stability of each candidate PS pixels will be estimated. The wrapped phase $\varphi_{x,i}$ of x^{th} pixel in i^{th} flattened and topographically corrected interferogram is given as [4]:

$$\varphi_{x,i} = W \left\{ \phi_{def,x,i} + \phi_{atmp,x,i} + \Delta\phi_{S,x,i} + \Delta\phi_{\theta,x,i} + \phi_{N,x,i} \right\} \quad (2)$$

where W is the wrapping operator $\{\cdot\}$, $\phi_{def,x,i}$ is the phase change due to the pixel motion in the satellite LOS, $\phi_{atmp,x,i}$ is the phase due to atmospheric delay, $\Delta\phi_{S,x,i}$ is the residual phase due to satellite orbit inaccuracies, $\Delta\phi_{\theta,x,i}$ is the phase due to look angle error and $\phi_{N,x,i}$ is the noise term introduced by the scattering effects [4]. The phase stability of the remaining pixels is examined using the following equation [4]:

$$\gamma_x = \frac{1}{N} \left| \sum_{i=1}^N \exp \left\{ \sqrt{-1 \left(\varphi_{x,i} - \tilde{\varphi}_{x,i} - \Delta\hat{\phi}_{\theta,x,i}^u \right)} \right\} \right| \quad (3)$$

For i^{th} interferogram image and x^{th} pixel, $\varphi_{x,i}$ is the wrapped phase, $\tilde{\varphi}_{x,i}$ is a wrapped estimate of the spatially correlated parts of $\varphi_{x,i}$, $\Delta\hat{\phi}_{\theta,x,i}^u$ is error phase and N is the number of interferograms. The pixels which satisfy the specified convergence rate are chosen as PS pixels.

Once PS pixels are chosen, their phase is unwrapped using phase unwrapping algorithms, details on which can be found in [4]. The final step is to calculate the spatially correlated look

angle (SCLA) error, which may be present due to the error used in the processing and simultaneously the master orbital and atmospheric errors, which makes the resultant phase value contain only the phase due to deformation. The outcome of this method results in the form of a PS velocity map which is a 1D-LOS map, showing the annual displacement.

IV. RESULTS AND DISCUSSION

In this section, results of StaMPS based PS-InSAR processing of the 13 Envisat ASAR SLC images of study area are presented with their interpretations. With 13 SLC images, 12 geocoded interferograms are generated as shown in Fig. 2. The values for D_A and γ_x adopted for the selection of PS candidates are 0.35 and 0.005 respectively and the pixel grid size is adjusted to 50 meters. Initially, more than 100000 PS candidates were selected based on the D_A value in each patch (an image scene is subdivided into six patches) which finally resulted in 5606 PS pixels. The processing resulted in generation of mean velocity in satellite LOS as shown in Fig. 4 (X and Y axis represents longitude and latitude respectively) in cold and warm colours with movements towards and away from the satellite respectively. The mean velocity of PS pixels in the figure ranges from (-17.9 to 26 mm/yr).

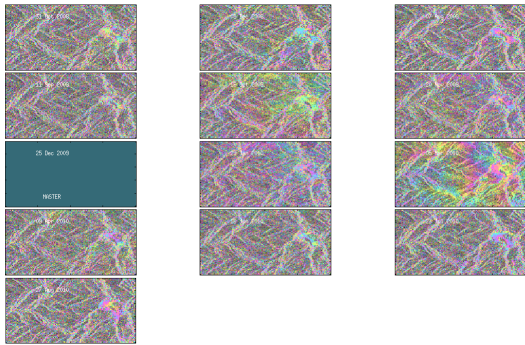


Fig. 2. Interferograms for the Nainital region. The image acquired on 25th December 2009 is chosen as the master image.

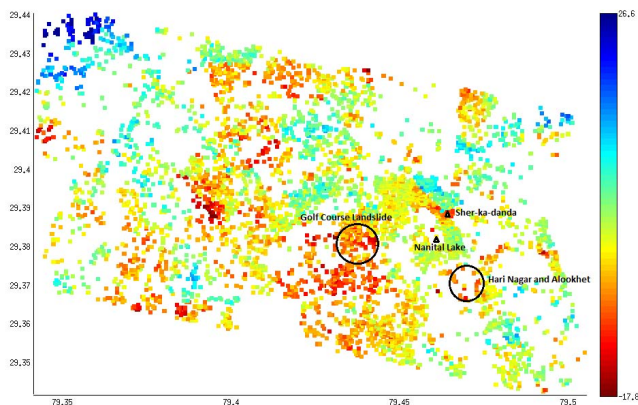


Fig. 3. PS velocity plot showing the PS pixel displacement in mm/year.

Fig. 3 shows PS velocity plots in the region highlighting that in the north-eastern and eastern part (Sher-ka-danda) of Nainital lake, is previously affected by subsidence and

landslide activities due to unstable slope. Although this region has remained stable in the last few years, sufficient numbers of PS pixels are extracted in this area with warm colours. In the south-western part of Nainital lake, an unstable area known as Golf course landslide is identified and circled in Fig. 3. From an earlier geotechnical assessment carried out in 2007-08 for the south-western part of the Golf course of Raj Bhawan complex in Nainital, the subsidence affected region of Golf course edge was reported to be located at the vertical cliff. Due to intersection joints, toppling of rock blocks and rock slide were observed at the edge of Golf course. It was also identified that the subsidence is reflected by ground fissures and seepage due to concealed drain. Although the location is densely vegetated, StaMPS processing successfully extracted PS pixels in that region and has been able to identify subsidence/landslip activities. It may be noted that on 17th and 18th September 2010, due to heavy rain, around 5 to 12 m subsidence was reported in the same region. This location is identified as critical zone after the geotechnical assessment and validation by StaMPS results. Further, subsidence in the vicinity of Nainital lake is also observed in the Fig. 3 with warm colours. This location around the Nainital lake known as Mallital which is the main shopping centre of the township. Further, on the south-eastern side of Nainital lake and along the Balia Nala, Hari Nagar and Aloo khet are two hilly areas which form a valley due to intense shearing. These two locations are also identified as landslide prone areas. Although PS densities in these two regions are slightly low but at least 10-12 pixels are identified with warm colours. The DMMC report [5] also marks landslide hazard zonation map (Fig. 4) after correlating various thematic maps prepared from satellite imagery and various data sources. The areas around Balia Nala and Golf course fall in the zone of high hazards which also validates the PS results.

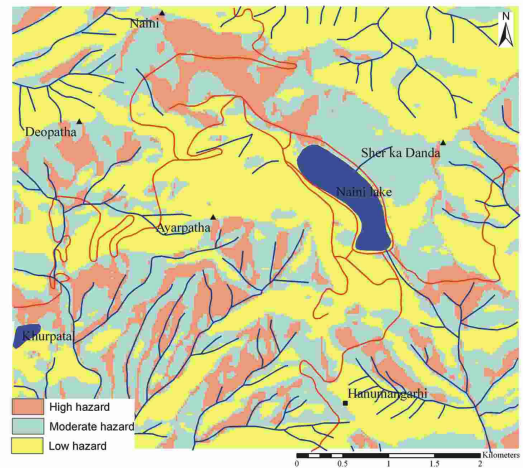


Fig. 4. Landslide hazard zonation map of the area around Nainital [5]

V. CONCLUSIONS AND FUTURE WORK

This study has investigated the efficacy of StaMPS based PS-InSAR approach for monitoring of critical landslides.

StaMPS successfully extracted the PS pixels in both urban and non-urban areas of the township and critical zones are identified. Further the generated time series deformation results are validated with the ground verification. In order to restore the unstable slopes around Nainital, authors are going to initiate a campaign mode geodetic survey using GNSS receivers. The time series analysis will be further improved if more image scenes of track 248 are available. In future, efficacy of small baseline subset (SBAS) InSAR approach will also be investigated in order to monitor land subsidence and landslide related activities around Nainital township.

VI. ACKNOWLEDGMENT

The authors are thankful to European Space Agency (ESA) for providing the ASAR images. Author are also grateful to Dr. Piyoosh Rautela, Executive Director, Disaster Management and Mitigation Center (DMMC) Uttarakhand for sharing the geological investigation report of the study site. The authors are thankful to NASA for providing the SRTM DEM and DORIS for providing the orbit files.

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